



Habib University

EE-172/CS-130/CE-222 Digital Logic and Design –
Spring 2026

Time = 50 minutes

Quiz 05 (CLO 4, Cog 5)

Max Points: 20

Question 1 [5 points]: Reduce the table using row reduction method and draw the reduced table.

Present State	Next state		Output	
	X=0	X=1	X=0	X=1
A	C	D	1	0
B	G	H	0	0
C	C	E	0	1
D	G	F	0	0
E	C	D	1	0
F	A	B	0	0
G	C	A	1	1
H	A	B	0	0

Present State	Next state		Output	
	X=0	X=1	X=0	X=1
A	C	B	1	0
B	C	F	0	0
C	C	A	0	1
F	A	B	0	0

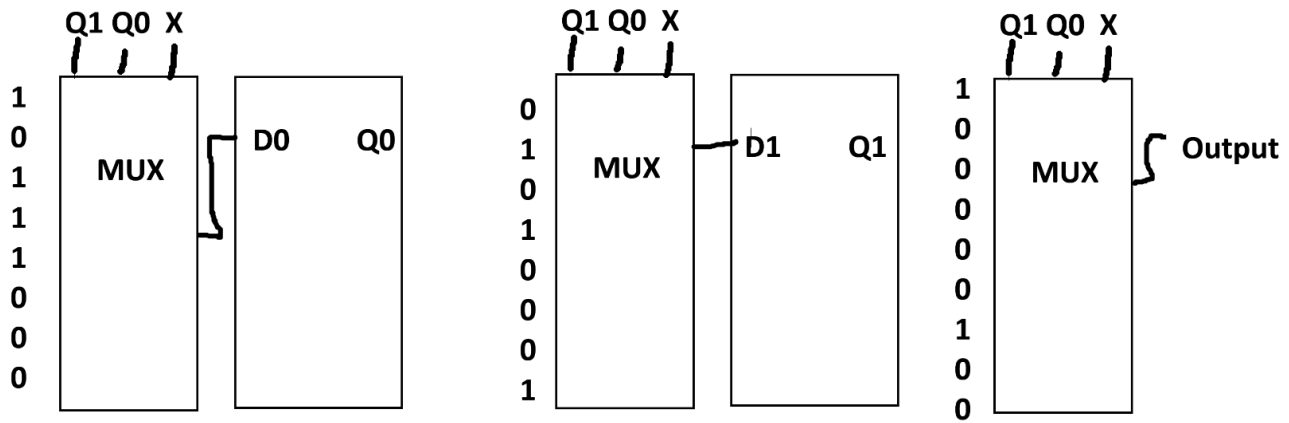
F=H, B=D, A=E, C = G

Question 2a) [5 points] Draw the circuit for the following state table using D Flip flops

$Q_1(t)$	$Q_0(t)$	X	$Q_1(t+1)$	$Q_0(t+1)$	Output
0	0	0	1	0	1
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	0	0	1
1	1	0	0	0	0
1	1	1	0	1	0

Any method to implement circuit accepted, easiest to simply use a 8 by 1 mux at each input and copy the table. Mux select lines should be $Q_1(t)$, $Q_0(t)$ and X.

Eq: $D_1 = Q_0' \cdot X' + Q_1' \cdot Q_0$, $D_0 = Q_2' \cdot X + Q_0 \cdot X$, $Y = Q_2' \cdot Q_1' \cdot X' + Q_2 \cdot Q_1' \cdot X$



Question 2b) [5 points]: Draw the circuit for **only Q₁** from the state table in 2a using JK flip flops.

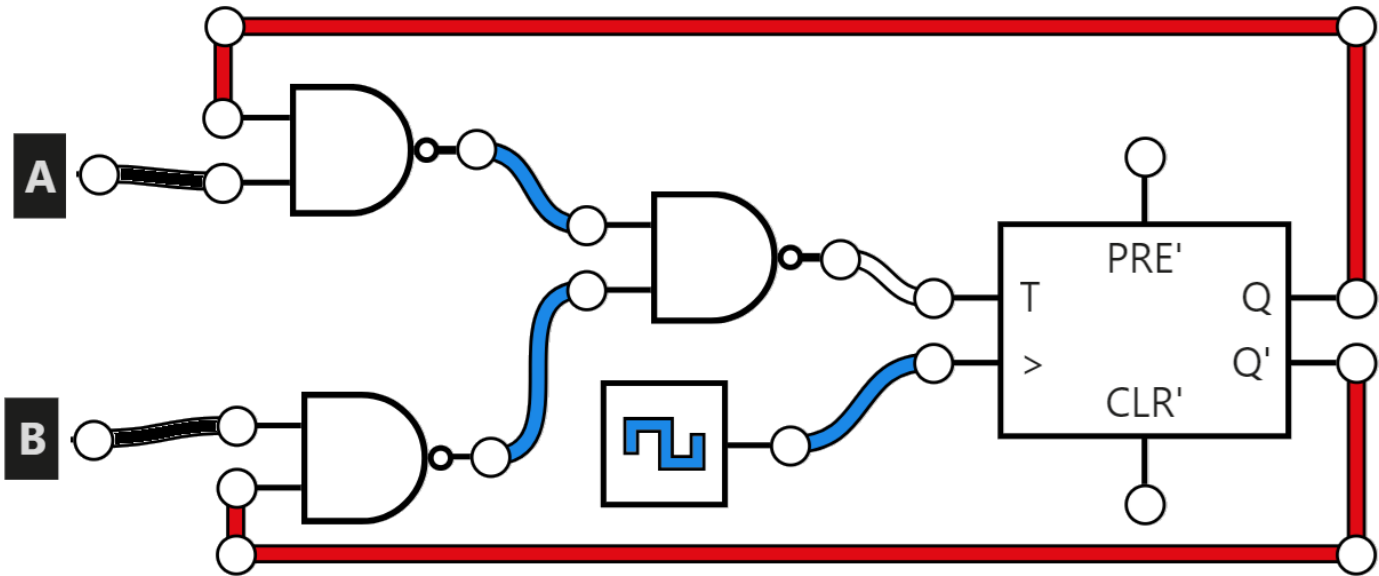
Solution:

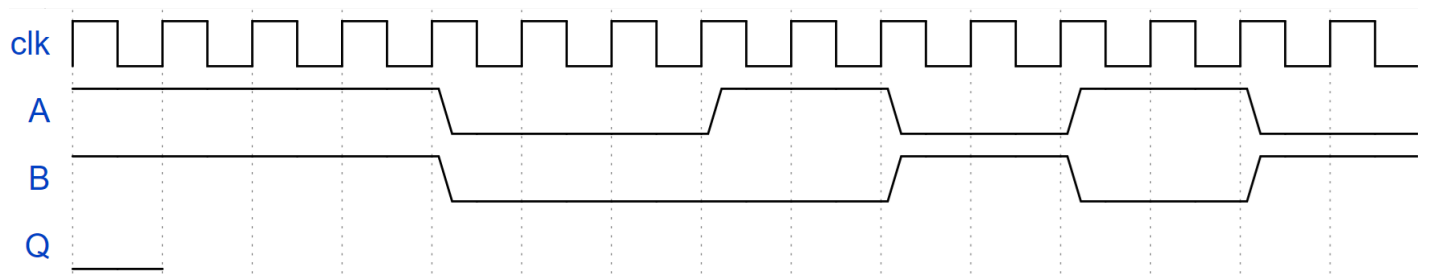
Q ₁ (t)	Q ₀ (t)	X	Q ₁ (t+1)	J	K
0	0	0	1	1	X
0	0	1	0	0	X
0	1	0	1	1	X
0	1	1	1	1	X
1	0	0	1	X	0
1	0	1	0	X	1
1	1	0	0	X	1
1	1	1	0	X	1

$J = \bar{X} + Q_0$, $K = X + Q_0$

Can be also drawn using MUX

Question 2c) [5 points]: Complete the following waveform for the circuit given:





Solution:

